Christian Oudard

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It turns out that if you really understand something well, you can teach a computer to do it. This has been a source of endless fascination for me throughout my life, ever since I implemented a chessboard on my TI-85 calculator in high school. For me, it all starts with gathering a precise understanding of what needs to happen. I'm a great communicator and I thrive in collaboration with others. In a team context, I'm often playing the role of a translator, who helps make technical realities legible for business-oriented roles, and represents business priorities for technologists.

I've written code for a wide variety of applications, including game development, graphics programming, server administration, full stack web development, cryptography, and financial analysis.

Skills

Some of my favorite tools and technologies are:

- NixOS Linux
- Git
- Vim
- Claude Code
- Cloud computing (GCP, AWS)
- Databases:
  - SQLite
  - Postgres
  - Redis
- Programming Languages:
  - Python
  - Rust
  - Go
  - Haskell
  - Lean
  - HTML / CSS / TypeScript
  - Shell script

Work History

Software Consultant

November 2017 – Present

Various small projects for private clients. I have helped people with their personal websites, bookkeeping systems for small businesses, cybersecurity audits, embedded software for scientific applications, oversight of junior programmers, and hiring due diligence, among other work.

Software Developer, RHS Financial

November 2017 – Present

Technologies: Python, CVXPY, Postgres, Pandas, GCP, FIX protocol

I have been working part-time with RHS Financial as the only software developer for quite a few years now, some of that as a minor partner in the firm. We have built a sophisticated portfolio optimization system using a convex optimizer. It rebalances client portfolios with a modern expected returns and covariance model, with automatic tax-loss harvesting. This system reduced the average client rebalance from two tedious hours (in Excel) to under ten minutes of work, including automated trade execution over the FIX protocol.

- https://rhsfinancial.com/

Senior Software Engineer, MobileCoin Inc.

May 2021 – October 2023

Technologies: `Rust`, `Python`, `gRPC`, `SQLite`, `SQLCipher`, Intel `SGX` enclaves, Elliptic Curve Cryptography

As a member of the core blockchain team, I worked with a Rust codebase doing high security cryptographic calculations. The most safety-critical portions of the code communicate in a decentralized consensus network, via secure enclaves. I implemented the client wallet software responsible for maintaining private keys and transaction history. Since MobileCoin is an end-to-end encrypted (E2EE) system, the client software requires an encrypted database for proper security. All the software I wrote here is open-source:

- `github.com/mobilecoinofficial/full-service`
- `github.com/mobilecoinfoundation/mobilecoin`

Engineer, Bonsai AI

March 2017 – October 2017

Technologies: `Python`, `Django`, `Docker`, `TensorFlow`

Bonsai AI built a reinforcement-learning platform and domain specific language, with the goal of making RL training and inference more accessible to non-expert developers. As a platform engineer, I found it fascinating to work directly with our ML researchers, helping to implement and scale RL algorithms. I learned a great deal about optimization theory and machine learning in general.

Data Engineer, Disqus

May 2016 – December 2016

Technologies: `Postgres`, `SQL`, `Python`, `Django`, `JavaScript`, `React`, `AWS`, `Apache Kafka`

I worked here on the data team, writing and managing batch database jobs to transform large data sets coming from their 500 million monthly unique users. Our three-person team was responsible for an analytics pipeline handling tens of millions of events per day. My role was fairly flexible, and I also helped with the Django backend and the React frontend.

- `https://disqus.com/`

Engineer, Infer Inc.

July 2015 – September 2015

Technologies: `Python`, `scikit-learn`, `CoffeeScript`

Infer was a company that helped to optimize sales pipelines via a Salesforce plugin. We used machine learning to help sales teams prioritize which contacts were most likely to convert into customers. My responsibility here was mainly to write an internal user interface for the data engineers to develop their ML models.

Senior Backend Engineer, TinyCo

September 2012 – January 2014

Technologies: `Python`, `Django`, `Tornado`, `MySQL`

TinyCo creates free-to-play mobile games, focusing on cute characters and time-locked game mechanics. I worked on the backend team. In collaboration with the mobile frontend team, we released several games a year, each one requiring its own backend code and database deployment. Here I coded complete rewrites of several major systems, such as the code deployment system and game configuration server. I also introduced better practices for unittesting and database migrations.

Engineer, Scribd Inc.

January 2012 – August 2012

Technologies: `Ruby on Rails`, `JavaScript`, `HTML/CSS`

As a full stack engineer, my responsibilities at Scribd included the Scribd Developer API, internal mobile client API, and the embedded document viewer.

Lead Software Developer, PolicyStat, LLC

October 2008 – January 2012

Technologies: `Python`, `Django`, `AWS`, `HTML/CSS`, `jQuery`, `Redis`, `MySQL`

I joined PolicyStat at a very early stage, when there was just the CEO, the CTO, and me. Over my time there, our client base grew from two hospitals to over a hundred. I did a little of everything here, including server administration, the Django backend, a responsive web frontend, and some help with pitch decks.

Web Developer, BluegrassNet Development

May 2008 – October 2008

Technologies: `PHP`, `MySQL`

My first job out of college was as a developer at a small custom web development shop, using PHP. We were making a trucking logistics platform for Tennessee Steel Haulers.

Software Engineering Intern, Flight Data Simulator Project

March 2008 – May 2008

Technologies: `Python`, `PyOpenGL`

Speed School Computer Science Department, University of Louisville

This internship involved writing 3D renderings of airplane landing approach paths.

Software Engineering Intern, CMTS Project

October 2006 – March 2008

Technologies: `C#`, `DirectX`

Speed School Mechanical Engineering Department, University of Louisville

This project, the "Concept Modeling Tool Suite", was a collaboration with the U.S. Army. It was aimed at designing soldier transport vehicles with reduced noise, vibration, and harshness characteristics. I wrote the 3D user interface to the vehicle modeling program we developed. The output was a vibration analysis produced by finite element modeling. Read all about the project in the dissertation of one of our researchers:

- `https://ir.library.louisville.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1816&context=etd`

Software Engineering Intern, Alcoa Louisville Foil Plant

August 2005 – August 2006

Technologies: `ASP.NET`, Microsoft Access

My work at the aluminum foil plant was to create data reporting pages for internal usage. I created an up-to-the-minute reporting system showing the readings from the foil mills, which roll sheet aluminum down to the thickness of kitchen foil.

Education

Bachelor of Computer Science

Fall 2004 – Spring 2008

University of Louisville, Speed School of Engineering